THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 1: **Patterns in Mathematics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A pattern adds the same number every time. Its first term is 9 and its 21st term is 169. The number added each time is: 1

(a) 8 **(b)** 9 **(c)** 10 **(d)** 7

2. A pattern adds 1, then 2, then 3, ... Starting at 7, the term after 18 additions is: 1

(a) 25 **(b)** 160 **(c)** 178 **(d)** 331

3. The sum $1^3 + 2^3 + \cdots + 11^3$ equals: 1

(a) 1331 **(b)** 66 **(c)** 132 **(d)** 4356

4. The 68th triangular number minus the 67th triangular number is: 1

(a) 67 **(b)** 69 **(c)** 1173 **(d)** 68

5. At a party, 25 people each hug every other person exactly once. Number of hugs: 1

(a) 300 **(b)** 325 **(c)** 312 **(d)** 600

6. If $1 + 2 + 3 + \cdots + n = 2016$, then n is: 1

(a) 63 **(b)** 32 **(c)** 64 **(d)** 62

7. The sum of the first 50 odd numbers is 2500. The sum of the odd numbers from 41 to 99 is: 1

(a) 2059 **(b)** 2480 **(c)** 2100 **(d)** 900

8. The difference between the 35th and the 34th square numbers is: 1

(a) 35 **(b)** 67 **(c)** 71 **(d)** 69

9. **Assertion (A):** In the Virahanka sequence $1, 2, 3, 5, 8, 13, \ldots$ the term after 13 is 21. 1
Reason (R): Each term after the first two is the sum of the two terms just before it.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** 36 is both a square number and a triangular number. 1
Reason (R): A number is a square number if it is a number multiplied by itself.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the sum of odd numbers from 27 to 65. 2

12. 19 cricket teams play a league in which every team plays every other team once. Find the number of matches. 2

13. The sum of the first n odd numbers is 3025. Find n and the 55th odd number. 2

14. Find the sum of odd numbers from 33 to 81. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a pattern the 1st term is 17, the 2nd is 25, and so on by adding 8. Which term of the pattern equals 425? Find also the term just before it. 3

16. Sums of consecutive odd numbers give squares. Find the sum of the odd numbers from 1 to 79. Then find the sum of the odd numbers from 81 to 99. 3

17. A tower is built in rows of cubes: row 1 has 1 cube, row 2 has 3, row 3 has 6, and so on (triangular numbers). How many cubes are in row 10, and how many in the first 6 rows together? 3

OR

In a pattern the 1st term is 4, the 2nd is 18, and so on by adding 14. Which term of the pattern equals 494? Find also the term just before it. 3

18. Sums of consecutive odd numbers give squares. Find the sum of the odd numbers from 1 to 89. Then find the sum of the odd numbers from 91 to 107. 3

OR

A tower is built in rows of cubes: row 1 has 1 cube, row 2 has 3, row 3 has 6, and so on (triangular numbers). How many cubes are in row 12, and how many in the first 6 rows together? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Meena is tiling a square floor. A 5×5 floor uses 25 tiles. To make a bigger square she adds an L-shaped border of tiles around two sides. 5

(i) How many tiles are in a 5×5 square floor?
(ii) How many tiles does she add to make it 6×6?
(iii) Which odd number is this?
(iv) Find the sum of the first 9 odd numbers. How many tiles must she add to go from 9×9 to 10×10?

20. In a pattern of dot figures, figure 1 has 6 dots and each next figure has 8 more dots than the one before. How many dots are in figure 15? Which figure has 158 dots? 5

OR

Two sequences: A is $9, 18, 27, 36, \ldots$ and B is $30, 35, 40, 45, \ldots$. Find the 13th term of each and decide which is larger and by how much. 5

END OF QUESTION PAPER — **Total: 40 Marks**